# Jagannath Narayanaswamy

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## EDUCATION

**George Washington University** **2024 - 2026**
*Master of Science in Data Analytics (SEAS Dean's Award)* *Washington, DC*

## PROFESSIONAL EXPERIENCE

**Center for Community Resilience** **Jan 2025 – Present**
*Data Scientist* *Washington, DC*

- **Led product decisions** for Economic Data Analytics Platform from zero-to-one, integrating Census Bureau, FRED, and HUD APIs serving 50+ researchers; automated workflows reducing manual processing by 80% through real-time orchestration and PostgreSQL caching.
- Engineered GPT-4 powered RAG system for natural language information retrieval, processing 500+ queries/week with 95% accuracy and enabling non-technical researchers to query complex demographic datasets instantly.
- **Collaborated with researchers and designers** to deploy React/Flask dashboard with ensemble ML forecasting (ARIMA, LSTM, Prophet), automating 5-year projections and reducing analyst workload by 60% with 92% prediction accuracy.

**Fitzhugh Mullan Institute for Health Workforce Equity, GWU** **Aug 2024 – Present**
*Research Specialist III (Data Scientist)* *Washington, DC*

- **Owned product direction** for U.S. Medicaid Primary Care Workforce Tracker—architected interactive platform processing multi-state healthcare datasets with geospatial analysis and real-time visualizations serving 100+ healthcare policymakers.
- Engineered scalable ETL data pipeline processing multi-state Medicaid datasets across 3,000+ counties with Python, implementing automated feature engineering, ML-based anomaly detection, and predictive analytics for provider shortage identification, reducing data quality issues by 95%.
- **Partnered with UX designers and stakeholders** to build responsive web dashboard with dynamic D3.js visualizations and customizable filters, enabling county-level workforce analysis with multi-dimensional filtering capabilities.

## TECHNICAL SKILLS

**LLMs & AI:** LangChain, LangGraph, GPT-4, Gemini, RAG Pipelines, Multi-Agent Systems, Prompt Engineering
**Languages & Frameworks:** Python, JavaScript, TypeScript, R, SQL, FastAPI, Flask, React, Next.js, Node.js
**ML/Data:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Time Series, NLP, PySpark
**Databases & Vector Stores:** FAISS, PostgreSQL, MongoDB, DuckDB
**Cloud & Tools:** AWS (EC2, S3, EMR), Google Cloud (Vertex AI), Docker, Git, n8n, Tableau, Power BI
**UI/UX & Design:** Responsive Design, Mobile-First Development, Cross-Device Optimization, Figma, User Research

## PROJECTS

**AdaptLearn AI — Multi-Agent Learning Product** (FastAPI, GPT-4, MemMachine) **Oct 2025**

- **Built zero-to-one with minimal resources under ambiguity**—architected multi-agent AI system with 3-layer RAG pipeline (Episodic, Profile, Knowledge Graph) using GPT-4 and MemMachine REST API, achieving 30% improvement in personalized learning outcomes.
- **Designed mobile-responsive UI optimized for multiple screen sizes**; engineered FastAPI backend with WebSocket orchestration, spaced repetition algorithms, emotional intelligence detection, and Redis caching delivering sub-3s response times.

**Promptly — AI Browser Extension Product** (Gemini Nano, Chrome Built-in AI) **Nov 2025**

- **Shipped end-to-end product with minimal resources**—built hybrid AI architecture with on-device Gemini Nano and cloud fallback, delivering 9 real-time analysis features including ArgumentArmor fallacy detection and multi-source fact-checking with 3s latency.
- **Crafted intuitive cross-device UI/UX** with responsive layouts for desktop and tablet; engineered specialized LLM prompt pipelines for sentiment analysis, toxicity detection, and debate mode with steelmanning logic.

**FinSight — Enterprise RBAC RAG Product** (FastAPI, LangChain, Chroma, DuckDB) **Sep 2025**

- Designed agentic SQL/RAG pipeline with LangChain orchestration and Cohere reranking, implementing classifier-driven query routing to handle mixed structured/unstructured queries with 90%+ accuracy.
- Built enterprise-grade RBAC system with per-role corpus isolation and CI/CD pipeline (PyTest + Playwright E2E), containerized for production deployment with zero-downtime updates.